Tangency, Discriminants, and Critical Values of a Projection

N-20220315

A parameter-dependent equation can have one, two, or several real solutions as its parameter changes. The number of solutions can change when roots collide or escape through the boundary of the admissible domain. In smooth geometry this is a critical-value phenomenon: a curve ceases to project regularly onto its parameter axis. The same rank condition lies behind constrained extrema.

§1.1 From a double root to a critical parameter

Consider

$$F(x,t) = x^2 - t = 0.$$

For $t < 0$ there are no real roots; at $t = 0$ the two branches meet; for $t > 0$ the roots are $x = \pm\sqrt{t}$. At the transition,

$$F(0,0) = 0, \qquad F_x(0,0) = 0,$$

while $F_t(0,0) = -1 \neq 0$. The zero set is a smooth parabola $t = x^2$; only its projection onto the t-axis has become singular at the vertex.

For a smooth family $F(x,t) = 0$, suppose $F_t \neq 0$. The implicit function theorem solves locally for $t = t(x)$, and differentiation gives

$$t'(x) = -\frac{F_x}{F_t}.$$

Therefore a critical point of the parameter projection satisfies $F = F_x = 0$. If in addition $F_{xx} \neq 0$, the local Taylor expansion has the form

$$t - t_0 = -\frac{F_{xx}(x_0,t_0)}{2F_t(x_0,t_0)}(x - x_0)^2 + o((x - x_0)^2).$$

The quadratic leading term explains why pairs of solutions generically appear or disappear with square-root scaling.

§1.2 The discriminant describes a concrete bifurcation

A more instructive example is

$$F(x,t) = x^3 - 3x + t.$$

The derivative $F_x = 3(x^2 - 1)$ vanishes at $x = \pm 1$. The corresponding parameter values follow from $F = 0$:

$$(x,t) = (-1,-2), \qquad (1,2).$$

For $-2 < t < 2$, the cubic has a local maximum of value $t + 2 > 0$ and a local minimum of value $t - 2 < 0$, hence three distinct real roots. For $|t| > 2$ it has exactly one real root. At $t = \pm 2$, two roots coalesce and a third remains simple.

The discriminant of $x^3 - 3x + t$ is

$$\Delta(t) = 108 - 27t^2 = 27(4 - t^2).$$

Its zeros locate precisely the two transition values. Algebraically, the discriminant vanishes when F and F_x share a root. Geometrically, those are the points at which the map from the solution curve to the parameter line develops a critical value.

For higher-degree polynomials, the *resultant* $\mathrm{Res}_x(F, F_x)$ generalizes this test. A zero resultant detects a common root over an algebraic closure, but whether that root is real and lies in the problem's domain must still be checked.

§1.3 Constrained extrema are the same rank failure

Let $g : \mathbb{R}^2 \to \mathbb{R}$ have nonzero gradient on the constraint curve $g(x, y) = 0$, and let h be the objective. The tangent space at a constraint point consists of vectors v satisfying

$$Dg(v) = \nabla g \cdot v = 0.$$

At a constrained extremum, the first variation of h vanishes along every such direction. Therefore ∇h is proportional to ∇g:

$$\nabla h = \lambda \nabla g.$$

Equivalently, the Jacobian of the map $(g, h) : \mathbb{R}^2 \to \mathbb{R}^2$ has rank below two:

$$\det \begin{pmatrix} g_x & g_y \\ h_x & h_y \end{pmatrix} = 0.$$

This is the constrained analogue of $F_x = 0$ for a parameter projection. In both cases, a map to the quantity being measured loses its local first-order freedom.

§1.4 Stationary points do not imply global extrema

Consider the objective $h(x, y) = x + y$ on the curve

$$g(x, y) = x^2 + y^3 - 2 = 0.$$

Since $\nabla g = (2x, 3y^2)$ never vanishes on this curve, every constrained stationary point satisfies the multiplier equations. Eliminating the multiplier gives

$$2x = 3y^2, \qquad x = \frac{3}{2}y^2,$$

and therefore

$$\frac{9}{4}y^4 + y^3 = 2.$$

There are exactly two real solutions for y: one positive and one below $-1/3$. To see this, differentiate the left-hand side:

$$\frac{d}{dy}\left(\frac{9}{4}y^4 + y^3\right) = 3y^2(3y + 1).$$

It decreases on $(-\infty, -1/3)$, increases on $(-1/3, \infty)$, and takes a value below 2 at its minimum. The two solutions thus yield the only constrained stationary points.

To classify them, work on the positive-x branch

$$x = \sqrt{2 - y^3}, \qquad H(y) = y + \sqrt{2 - y^3}.$$

At a stationary point $x = 3y^2/2$, one obtains

$$H''(y) = -\frac{2(3y + 1)}{3y^2}.$$

The negative solution gives $H'' > 0$, a local minimum; the positive one gives $H'' < 0$, a local maximum. The negative-x branch has no stationary points.

Neither is a global extremum on the entire constraint curve. Taking $y = -R$ with $R \to \infty$ gives

$$x = \pm\sqrt{2 + R^3}, \qquad h(x, y) = \pm\sqrt{2 + R^3} - R,$$

which tends to $+\infty$ and $-\infty$ along the two branches. The feasible set is noncompact, so local stationarity says nothing about existence of a global maximum or minimum.

A discriminant tells us where algebraic branches collide; the implicit function theorem explains their local geometry; Lagrange multipliers detect the same loss of rank along a constraint. The distinctions among local, global, real, and admissible solutions determine which conclusions actually follow from the equations.